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(54) **WEED AND LINE CUTTER INSTALLATION
TOOL**

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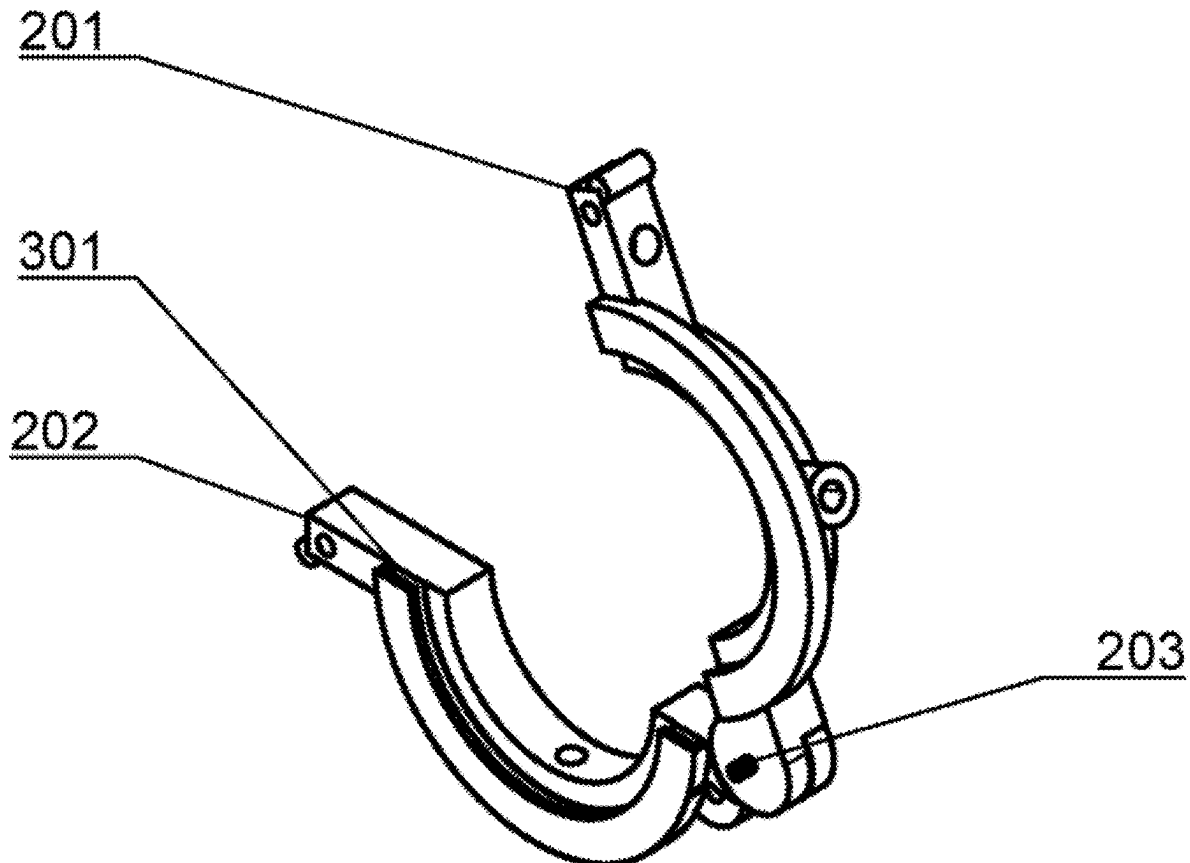
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(57)

ABSTRACT

The present invention relates to a novel apparatus and method for installing split line cutters on marine propeller shafts. The apparatus features a clamshell design comprising an upper and lower half, each with an inner cavity configured to accommodate a portion of the split line cutter. A hinge mechanism connects these halves, facilitating easy opening and closing around the propeller shaft. Key to this invention is a captive screw retaining mechanism within the clamshell, designed to hold screws in place even when unfastened, thereby simplifying the installation process and preventing loss of components. The apparatus includes a screwdriver opening aligned with a bolt cavity, allowing for engagement of captive screws. Additionally, the invention features temporary closure mechanisms, for secure handling during installation, and a retaining hole for attaching a line. This invention offers significant improvements in the installation of line cutters, emphasizing safety, efficiency, and ease of use.



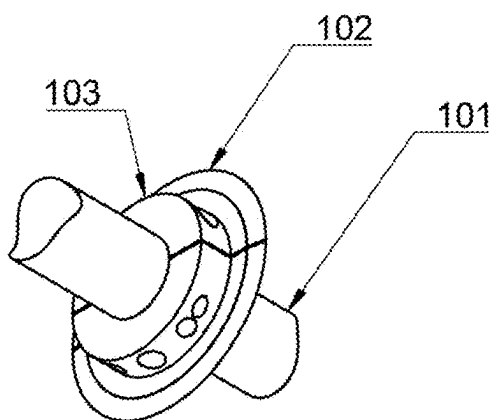


Fig. 1A

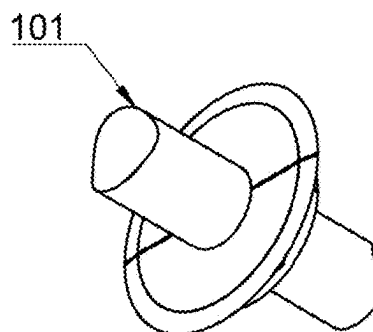


Fig. 1B

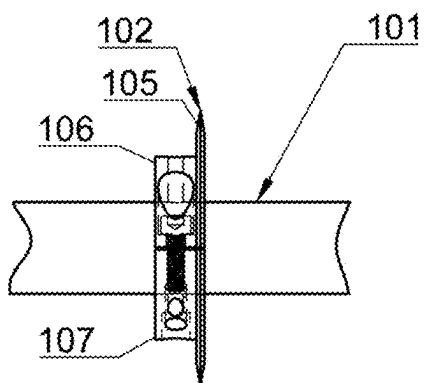


Fig. 1C

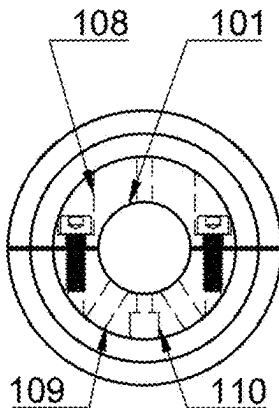


Fig. 1D

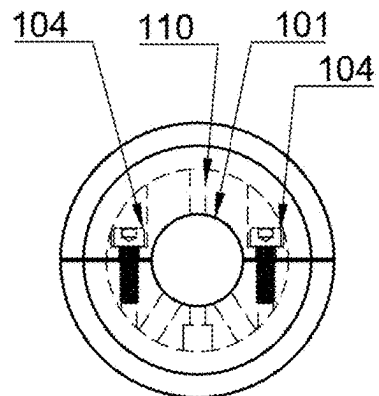


Fig. 1E

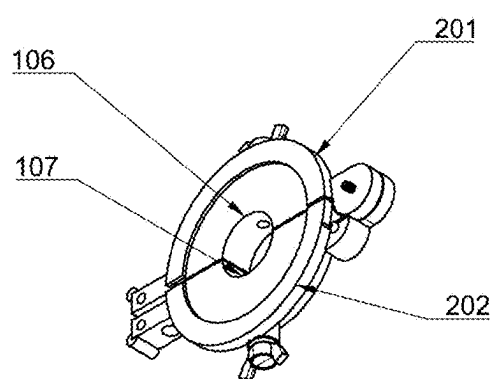


Fig. 2A

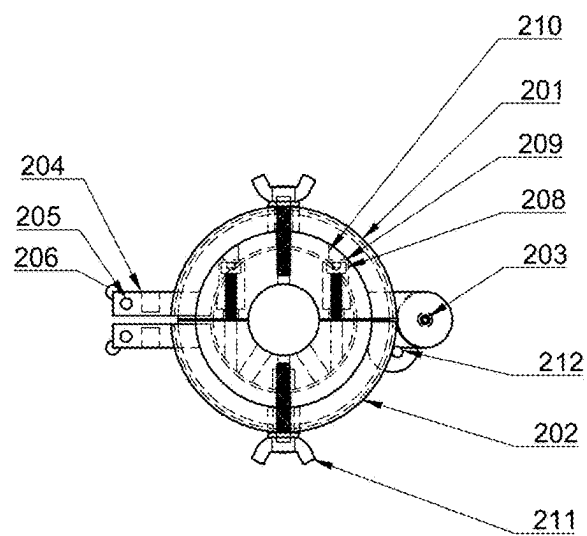


Fig. 2B

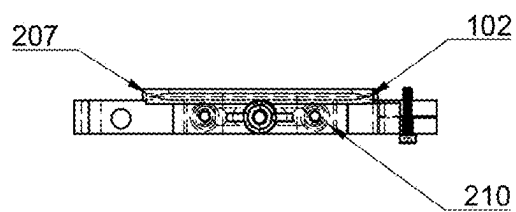


Fig. 2C

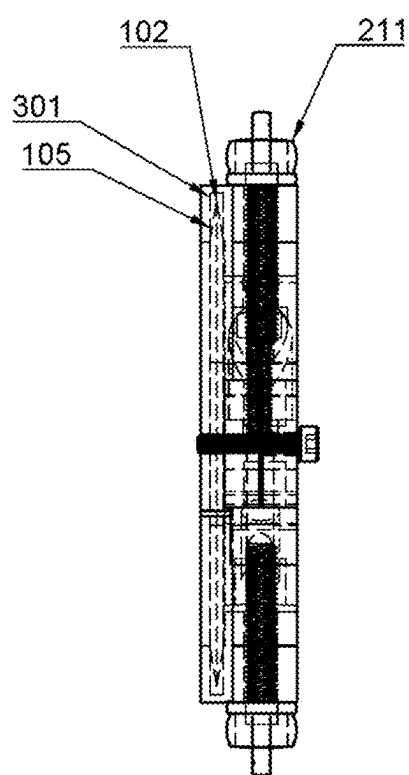


FIG. 3

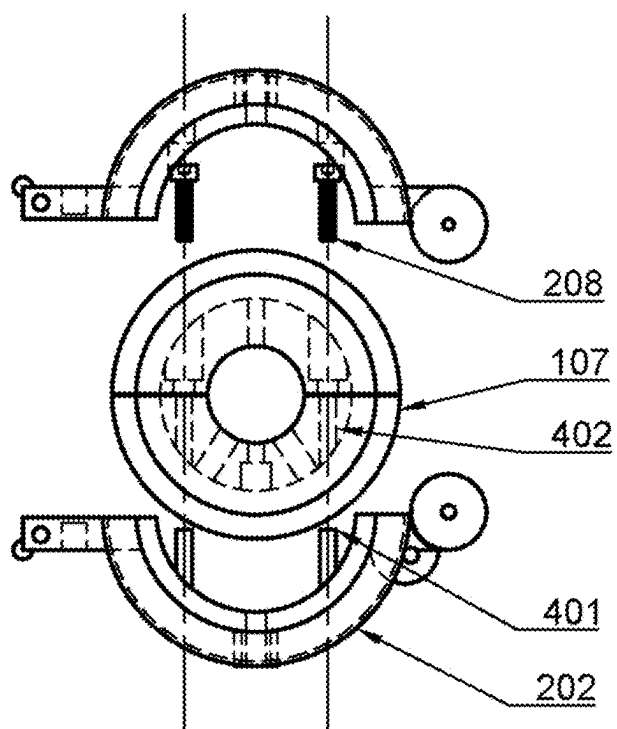


FIG. 4

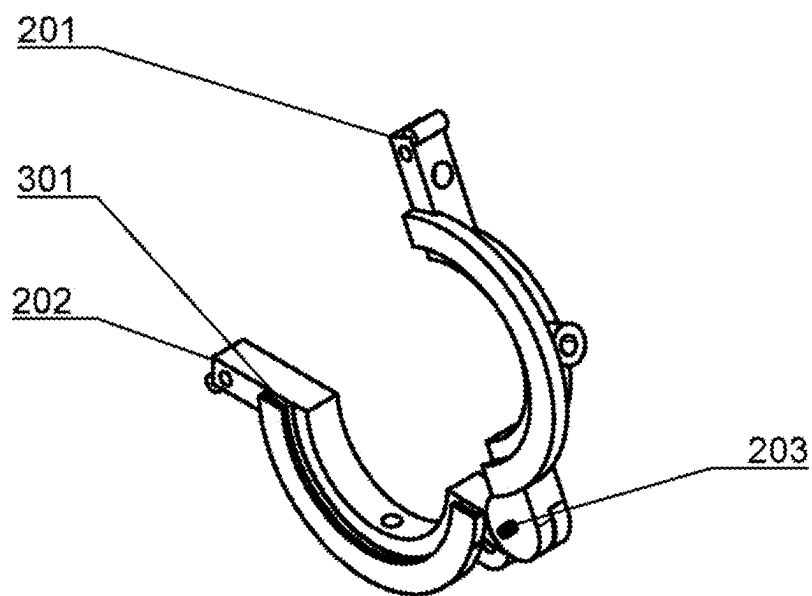


FIG. 5

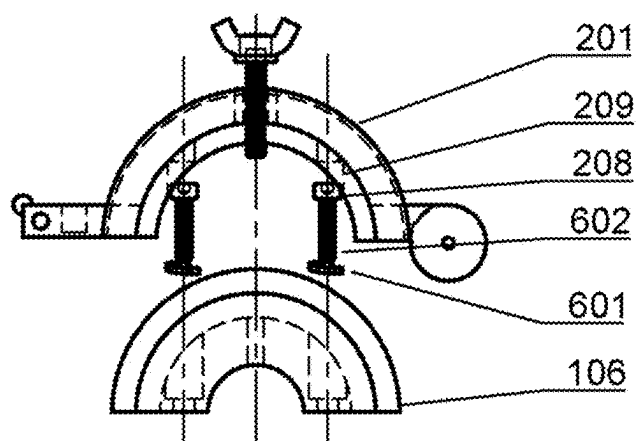


FIG. 6

WEED AND LINE CUTTER INSTALLATION TOOL

REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation application of and claims priority under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application No. 63/620,804 filed Jan. 13, 2024 titled “Weed and Line Cutter Installation Tool”.

FIELD OF THE INVENTION

[0002] The present invention relates to weed and line cutters installed on recreational and commercial marine vessels and in particular to tools used to install weed and line cutters on recreational marine vessels.

BACKGROUND OF THE INVENTION

[0003] Line and weed cutters are sharp instruments installed on the propeller shaft of commercial or recreational marine vessels. When these vessels traverse inland and ocean waterways, long strands of material may wrap around the propeller shaft and stall the engine. This material could be man-made material such as rope or fishing line, or it could be natural material such as weeds.

[0004] To prevent such an occurrence, a line and weed cutter (referred to as simply a “line cutter”) may be installed on the propeller shaft. There are many different designs of these cutters, but they all share the use of a sharp edge that rotates with the propeller shaft. It uses the power of the rotating shaft to slice through anything that begins to wrap around the shaft.

[0005] The line cutter may be for example a simple disc that is concentric with the shaft itself. The edges of the disc are sharpened to make a circular knife. If a rope, line, or weed begins to wrap around the shaft, the pressure of the rope on the shaft will press against the knife-edge and sever the rope, line, or weed.

[0006] The installation of these line cutters is normally done with the boat out of the water. Some line cutters require the removal of the propeller, but some line cutters are split and designed to allow installation without removing the propeller.

[0007] Because the line cutters by design have sharp knife-like edges, installing them requires great care to avoid cuts or nicks. The sharp edge may have some protective barrier that can be removed during or after installation. This protective barrier may be for example tape or rubber placed over the sharp edge. This protective barrier is not strong enough to prevent damage if the line cutter is dropped, and does not assist with alignment of the cutter if it is of the split type.

[0008] While the split type line cutters may be installed without removing the propeller, the presence of the sharp edge makes installing them a task that requires care and dexterity. The two sharp halves need to be held together while the bolts are threaded into the cutters.

[0009] It is typical that vessels are removed from the water at a commercial marina, which entails cost of removing and replacing the vessel which would typically be much higher than the cost of the line cutter itself. For this reason the installation of the line cutter is typically done when the boat is removed for other reasons, such as the owner will not be using the boat over the winter.

[0010] While the split type of line cutters can be installed underwater, saving the cost of removing the boat from the water, this is not trivial in practice. The protective barrier can be difficult to remove underwater with gloves, as it requires a high degree of dexterity. This entails removing the protective barrier and exposing the sharp edge above water. If part of the line cutter or the bolts are dropped it will be difficult or impossible to recover. Divers may have limited dexterity, especially if wearing gloves due to the cold water.

[0011] If a line cutter is being installed underwater by a snorkeler or free-diver, they have limited time that can be spent with each install step. With a typical split line cutter there is a significant risk of the cutter falling off the shaft, as until the bolts are secured the person installing the line cutter must hold it on the shaft. For these reasons even the split type line cutters are normally installed while the vessel is out of the water.

[0012] When installing a line cutter with the boat out of the water, a dropped line cutter will not be lost. But the sharp edge will be easily damaged if dropped onto hard ground. The protective layer installed on the sharp edge by the manufacturer is often only to protect the user from the sharp edge, and is not strong enough to protect the sharp edge from the weight of the line cutter being dropped.

[0013] What is needed is a tool to install the line cutter and hold captive all parts of the cutter, which protects the sharp edge from both damage and accidental contact, and the tool is only removed once the installation is complete. The tool needs to be self aligning and self holding to allow installation by a diver underwater.

OBJECT OF THE INVENTION

[0014] It is an object of this invention to: protect the sharp edge of the line cutter from damage when the line cutter is dropped onto hard ground such as found in typical boat yards, typically asphalt, gravel, or concrete; securely hold a split line cutter onto the shaft during the installation process; and to hold captive parts of the line cutter including the bolts until the installation is complete.

[0015] Still other objects and advantages of the invention will in part be obvious and will in part be apparent from the specification and drawings.

SUMMARY OF THE INVENTION

[0016] This invention relates to a split line cutter installation tool designed for mounting line cutters on the shafts of marine propellers on vessels. The tool simplifies the installation process, enhances safety, and ensures a secure fit of the line cutter. The tool has a clamshell design, and is made of a material suitable for the marine environment, while also being a material that is low-cost and suitable for packaging the line cutter itself in.

[0017] Key features of the tool include its split design, a captive screw retaining mechanism with driver guide, an alignment and securing system, the construction from suitable materials for the environment and commercial relevance, and the improved ergonomic and safe handling of the installation tool.

[0018] The split design of the tool comprises two halves, often referred to as a clamshell design, which can be opened and closed around the propeller shaft. This split design

allows for easy placement and removal of the tool without the need to disassemble other parts of the propeller assembly.

[0019] A captive screw retaining mechanism with driver guide is a key feature of this tool. The screws necessary for securing the line cutter to the shaft are retained within the tool itself, even when loosened. This design prevents the loss of screws during installation or removal in challenging marine environments. Openings in the tool provide a guide for the installation screwdriver, allowing installation with no visibility.

[0020] The alignment and securing system aligns the line cutter correctly on the shaft, including aligning bolt holes in each half of the shaft. Once aligned, the tool securely holds the line cutter in place while it is being fastened, reducing the risk of misalignment or slippage. An optional safety strap can be temporarily attached to the propeller shaft, which holds the installation tool and line cutter in case it is dropped during installation. The aligning and securing system also holds the line cutter in the tool itself, to ensure the line cutter is not dropped when opening and closing the tool.

[0021] The material and durability is important for both the marine environment and the commercial use. The tool may be sold with the line cutter, which requires consideration for the cost of the installation tool. As the user may wish to reuse the tool for future removal or storage of the cutter, the tool must be constructed of a material which meets both the cost requirements, along with withstanding typical marine environments.

[0022] The tool is designed with ergonomics in mind, making it easy and safe to handle, even on wet or slippery decks. This feature reduces the risk of accidents during the installation process, which may result in damage or loss of the tool and line cutter, or harm to the installer.

[0023] The invention accordingly comprises the several steps and the relation of one or more of such steps with respect to each of the others, and the apparatus embodying features of construction, combinations of elements and arrangement of parts that are adapted to affect such steps, all is exemplified in the following detailed disclosure, and the scope of the invention will be indicated in the claims.

BRIEF DESCRIPTION OF DRAWINGS

[0024] For a more complete understanding of the invention, reference is made to the following description and accompanying drawings, in which:

[0025] FIG. 1A Shows a line cutter installed on a propeller shaft;

[0026] FIG. 1B shows a view of the line cutter installed on a propeller shaft, the view being rotated approximately 180 degrees from FIG. 1A;

[0027] FIG. 1C shows a side view of the line cutter installation showing the two halves;

[0028] FIG. 1D shows the needed balancing holes;

[0029] FIG. 1E shows the mounting bolts;

[0030] FIG. 2A shows a top view of one embodiment of the line cutter installation tool;

[0031] FIG. 2B shows an orthogonal view of one embodiment of the line cutter installation tool;

[0032] FIG. 2C shows a side view of one embodiment of the line cutter installation tool;

[0033] FIG. 3 shows a side view of one embodiment showing a cavity for protecting the sharp edge of the line cutter;

[0034] FIG. 4 shows an exploded view showing one embodiment of the retaining mechanism using a plastic plug;

[0035] FIG. 5 shows a view of an embodiment of the installation tool without the line cutter installed; and

[0036] FIG. 6 shows an exploded view of the captive bolt retaining mechanism including a lock washer and adhesive.

DETAILED DESCRIPTION OF THE INVENTION

[0037] FIG. 1A-1E show different views of a typical split line cutter installed on a propeller shaft **101**. FIG. 1A shows the installation on the propeller shaft **101**, and FIG. 1B shows a view of this installation from the other side of the propeller shaft **101**. FIG. 1C shows that the line cutter is split into an upper **106** and lower **107** portion. The line cutter has a sharp edge **102** with an angled section **105** to provide support for the sharp edge **102**, and allow for future sharpening of sharp edge **102**. FIG. 1A shows that the line cutter has a larger hub portion **103** which allows mounting bolts shown in FIG. 1E as **104** to secure the upper **106** and lower **107** portion together, clamping around the propeller shaft **101**.

[0038] The mounting bolts **104** will typically be a socket head type, which use a hex screwdriver (often referred to as an Allen key). The mounting bolts **104** will have some form of locking to prevent the bolts from backing out due to vibration. This might include a locking material on the bolt thread such as a glue or rubber, or mechanical means to hold the bolt secure such as a lock washer, a locking cable, or another bolt that threads into or against the mounting bolts **104**.

[0039] The mounting bolt countersink **108** causes the upper **106** and lower **107** half of the split line cutter to be different weights and thus unbalanced, which may lead to vibrations at higher propeller speeds. As shown in FIG. 1D to prevent this balancing holes **109** can be drilled in the lower **107** half of the split line cutter. Additionally threaded holes **110** may be added to the upper **106** and lower **107** portions for a locking set screw to be used to further lock the line cutter onto the shaft.

[0040] FIG. 2A-FIG. 2C show different views of one embodiment of the installation tool for the split line cutter. FIG. 2A shows how the upper **106** and lower **107** portion of the split line cutter present in each part of the installation tool. Further figures will show the tool alone (without upper **106** and lower **107** portion present) along with the tool used during the installation itself.

[0041] FIG. 2B shows the tool has two clamshell halves, an upper clamshell half **201** and a lower clamshell half **202**. A hinge point **203** allows the two halves to open. A retaining mechanism **211** holds each half of the line cutter in the clamshell. The retaining mechanism **211** in this embodiment is a wingbolt, but other mechanisms such as friction fit of the clamshells, adhesives, thumbscrews, bolts, and plastic plugs are all similarly suitable for the purpose of retaining each of the upper **106** and lower **107** sections of the line cutter in the upper clamshell half **201** and lower clamshell half **202** section of the clamshell.

[0042] During installation, the upper clamshell half **201** and a lower clamshell half **202** need to be held together temporarily using a temporary closure mechanism. Several embodiments of this mechanism are demonstrated in FIG. 2B, and the tool may include only a single or multiple types

of temporarily closure mechanisms. One embodiment is to use a magnet **204** which are installed on both the upper clamshell half **201** and a lower clamshell half **202**. The magnets are oriented such they will have a strong attractive force, and this is especially effective if a rare-earth magnet such as a neodymium magnet is used. The user can close the tool around the propeller drive shaft, and it will securely stay shut to complete the installation process.

[0043] Another embodiment of the temporary closure mechanism is the holes **205** present in the upper clamshell half **201** and a lower clamshell half **202**. These holes allow a suitably shaped fork or plug to be inserted, which will hold the two sides closed. The holes may also be used to pass a string or line through, which can be used to either hold the tool closed or to secure the tool to prevent it from being dropped. Finally the protrusion **206** is designed to be held by hand, which allows one to hold the mechanism closed with a single hand. This same protrusion **206** also serves as a stop for a line, string, rubber band, or zip-tie which could be passed over the end of the tool. This will allow the tool to be securely closed, and the protrusion **206** prevents the line, string, rubber band, or zip-tie from falling off the end of the tool.

[0044] During installation, there is a risk the tool is dropped. This is especially the case if the line cutter is being installed underwater. For this purpose a retaining hole **212** is included. A line, chain, string, or rope can be attached to the retaining hole **212**. Because the line cutter halves are secure to the tool using the retaining mechanism **211**, this single retaining hole **212** allows both the tool and line cutter to be secured from the attached line, chain, string, or rope. During the installation process the lanyard or retaining hole **212** allows the installation tool to be temporarily attached to the vessel, by being clipped to any conveniently available protrusion from the vessel. This could include for example a hook or wire attached to the driveshaft itself.

[0045] The inclusion of the retaining hole **212** allows the installation of the line cutter to occur when the process must be interrupted, such as would occur if the installation is happening using only a mask and snorkel. In this example the installation tool may first be attached to a lanyard by the retaining hole **212** which is passed over a hand, to ensure the item is not dropped. The lanyard is then transferred to the vessel itself. Once the installation is complete and the tool is to be removed, the lanyard is then removed from the vessel.

[0046] During installation, the upper clamshell half **201** and a lower clamshell half **202** are closed around a propeller drive shaft. the retaining mechanism holds the two halves closed as discussed. Previously in the installed line cutter from FIG. 1C, the mounting bolts **104** can be seen securing the line cutter to the drive shaft. In FIG. 2B, the mounting bolts are not yet installed, and are marked as the captive mounting bolts **208**. The bolts are held captive by a bolt cavity **209**, which is part of the upper clamshell half **201**.

[0047] The construction of the hinge **203** ensures that on closure the two halves are correctly aligned, and the captive mounting bolts **208** will thread into the lower **107** section of the line cutter. This will clamp the upper **106** and lower **107** sections of the line cutter onto the drive shaft. To allow tightening of the captive mounting bolts **208**, a screwdriver opening **210** allows the appropriate screwdriver to pass through and engage with the hex captive mounting bolts **208**. Typically this would be a hex or ball-head hex screwdriver

as used by captive screws which are shown in FIG. 2B and FIG. 2C, but those skilled in the art will recognize there are many forms of screwdrivers and bolt heads.

[0048] The screwdriver opening **210** in combination with the bolt cavity **209** also ensure that the bolt can be screwed in without visually seeing the screw head. The recessed screwdriver opening **210** can be made of the size such a screwdriver will fit through and engage the captive mounting bolts **208**, and this can be felt as a positive engagement when using a hand tool. The screwdriver opening **210** may have additional design for improved tactile feedback, such as having a sloped funnel shape to guide the screwdriver into the screwdriver opening **210**, or having a built-up cylinder around the screwdriver opening **210** to raise it above the height of the body of the tool.

[0049] Once the captive mounting bolts **208** hold the upper **106** and lower **107** half of the split line cutter together, the installation tool can be removed. This can be done once the mounting bolts are fully tightened and the line cutter cannot move, or can be done before they are fully tightened. Depending on the installation for example it may be easier to access bolts from certain angles, and it is desired to have the ability to move the line cutter on the propeller shaft after the installation tool is removed. The installation tool is removed by removing the retaining mechanism **211**, in the embodiment of FIG. 2B this would mean removing the wingbolt.

[0050] FIG. 3 shows another side view of the tool, with the sharp edge **102** of the line cutter marked. The clamshell inner cavity **301** is designed to protect the sharp edge **102**, by providing some clearance from the sharp edge **102** and clamshell inner cavity **301**. The clamshell inner cavity **301** may be designed to rest on the angled section **105** of the line cutter, which in combination with a choice of a slightly flexible material for the tool provides a friction fit of the half of the line cutter in the clamshell. This may be used in place of or in addition to the previous embodiment of a retaining mechanism **211** that held the line cutter in the clamshell with a threaded wingbolt.

[0051] Another embodiment of a retaining mechanism is shown in FIG. 4, where a plastic plug **401** that may be a separate part that is inserted into a suitable hole in the lower clamshell half **202**, or may be made an integral part of the lower clamshell half **202**, extends into the lower **107** half of the line cutter bolt hole **402**. In this design as the captive mounting bolts **208** are tightened, the plastic plug **401** is forced out of the lower **107** half of the line cutter. If the plastic plug **401** is a separate item inserted into the lower clamshell **202** this will ensure that the retaining mechanism is automatically released once the captive mounting bolts **208** begin to be tightened. The two clamshell halves will still be held together by the temporary retaining mechanism. If the plastic plug **401** is an integral part of the lower clamshell half **202**, this will also force open the lower clamshell half **202**, again releasing the tool from the line cutter, while also releasing the temporary retaining mechanism. These provide two different user experiences and may be further optimized for different use-cases and installation procedures.

[0052] A view of the installation tool without the line cutter installed is given in FIG. 5. Here again the upper clamshell half **201** and lower clamshell half **202** are joined by a hinge **203**. The clamshell inner cavity **301** is visible.

[0053] The material for the installation tool will depend on the specific use-case and commercial marketing plan. If the

tool is sold with a line cutter, a low-cost and lightweight material such as a plastic may be preferred. Such a material provides a simple method of safely shipping the line cutter to the end customer, as the sharp edge is prevented from being exposed. The sharp edge is also protected from damage in the event the line cutter is dropped, and there is no risk of the line cutter being damaged when being transferred to a different installation tool. The installation tool can also be made from techniques such as 3D printing, which allows rapid stocking of tools with varying dimensions to suit line cutters coming from many vendors.

[0054] If the installation tool is to be used commercially, a more robust material such as stainless steel may be preferred instead of a plastic material. This has advantages of a more precise and rigid design, which will improve the ability of the tool to align the mounting bolt holes. This would be considerably more expensive than a simple plastic model, and would not be suitable for use as a low-cost shipping container.

[0055] The installation tool may also have transparent sections, using a transparent material or openings along the tool. This can assist with the installation by providing a method of ensuring the line cutter is correctly aligned before tightening.

[0056] Further modifications to the design to assist with alignment can be made to ensure the line cutter is a correct distance from the other sections of the driveshaft, such as a bearing or hub. This can be done with markings on the installation tool, or by modifying the design of the installation tool to include spacers or open sections.

[0057] FIG. 6 shows an exploded view of the upper clamshell half **201** with the upper **106** cutter section, and the captive mounting bolts **208**. The design of the cutter with captive bolt cavity **209** also makes it simple to add the required bolt locking mechanisms that will prevent the captive mounting bolts **208** from coming loose due to vibration or rotation of the line cutter. A lock washer **601** is added to the captive mounting bolts **208**, and because the captive mounting bolts **208** are held securely between the captive bolt cavity **209** and upper **106** cutter section, these lock washer **601** cannot be accidentally dropped during shipping or installation. Other bolt locking mechanisms are also compatible with the captive bolt cavity **209**, such as using a locking fluid **602**, which may be for example some form of plastic, rubber, or adhesive. Normally the locking fluid **602** would be placed on the bolt only before installation of the bolt, but because the bolt is protected within the captive bolt cavity **209** the locking fluid **602** can be installed by the manufacturer. This demonstrates another advantage of the line cutter installation tool holding all parts of the line cutter together until it is installed.

[0058] It will thus be seen that the objects set forth above, among those made apparent from the preceding description, are efficiently attained and, because certain changes may be made in carrying out the above method and in the construction(s) set forth without departing from the spirit and scope of the invention, it is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

[0059] It is also to be understood that the following claims are intended to cover all of the generic and specific features of the invention herein described and all statements of the scope of the invention which, as a matter of language, might be said to fall therebetween.

1. An installation tool for mounting a split line cutter onto a marine propeller shaft, the split line cutter having an upper portion, a lower portion, and at least one mounting bolt that secures the upper and lower portions around the propeller shaft, the installation tool comprising:

- (a) a first clamshell half configured to receive at least a portion of the split line cutter, the first clamshell half including a first inner cavity sized and shaped to protect a sharpened edge of the split line cutter;
- (b) a second clamshell half hingedly coupled to the first clamshell half, the second clamshell half including a second inner cavity sized and shaped to protect another portion of the split line cutter;
- (c) a retaining mechanism disposed on at least one of the first clamshell half or the second clamshell half, the retaining mechanism being configured to hold the split line cutter portions within the respective inner cavities when the installation tool is in a closed position;
- (d) one or more bolt cavities formed in at least one of the clamshell halves, each bolt cavity being configured to hold the mounting bolt in a captive position, whereby a driver may engage the mounting bolt through a driver opening in the installation tool; and,
- (e) a temporary closure mechanism that provides a temporary locking force that retains the installation tool in the closed position,

wherein the installation tool is adapted to (i) secure the split line cutter during alignment on the marine propeller shaft, (ii) protect the sharpened edge of the split line cutter, and (iii) allow a user to fasten the mounting bolt without directly contacting the sharpened edge.

2. The installation tool of claim 1, wherein the temporary closure mechanism is a magnetic closure disposed on each of the first and second clamshell halves, the magnetic closure providing a temporary locking force that retains the installation tool in the closed position around the propeller shaft.

3. The installation tool of claim 1, wherein the retaining mechanism comprises a removable plug that extends into a bolt hole of one of the split line cutter portions, and is automatically forced out of the bolt hole upon tightening of the mounting bolt.

4. The installation tool of claim 1, wherein at least one of the first clamshell half or the second clamshell half includes a retaining hole adapted to receive a tether, lanyard, or other securing device, thereby preventing accidental loss of the installation tool in an underwater environment.

5. The installation tool of claim 1, wherein the driver opening is shaped to guide a screwdriver or hex key into engagement with the mounting bolt without requiring direct visual alignment, providing tactile feedback to the user as the mounting bolt is tightened.

6. The installation tool of claim 1, wherein at least one of the first clamshell half or the second clamshell half is formed from a polymeric material that is sufficiently rigid to protect the sharpened edge of the split line cutter and sufficiently low-cost to serve as both packaging and installation aid for the line cutter.

7. The installation tool of claim 1, further comprising an optional safety strap configured to temporarily attach the tool to the propeller shaft, preventing the tool and enclosed line cutter portions from being dropped during installation.

8. The installation tool of claim **1**, further comprising the hinged connection being of sufficient rigidity to ensure alignment of the split line cutter.

9. A method for installing a split line cutter onto a marine propeller shaft using an installation tool having first and second clamshell halves hingedly coupled, at least one retaining mechanism, and at least one bolt cavity for a mounting bolt, the method comprising the steps of:

- a. closing the clamshell halves around the propeller shaft, thereby aligning the split line cutter portions to clamp onto the shaft;
- b. inserting a driver through at least one driver opening in the installation tool, the driver engaging the mounting bolt residing in the bolt cavity of the installation tool;
- c. fastening the mounting bolt to secure the upper portion of the split line cutter to the lower portion on the propeller shaft; and
- d. removing the installation tool by releasing the retaining mechanism after the mounting bolt is sufficiently tightened,

whereby the sharpened edge of the split line cutter is protected and the mounting bolts are retained against loss during installation.

10. The method of claim **9**, wherein a magnetic closure holds the clamshell halves together during step (c), further comprising the step of detaching the magnetic closure by manually applying force to separate the first and second clamshell halves.

11. The method of claim **8**, further comprising the step of attaching a tether or lanyard to the installation tool prior to submerging the tool, such that the tool and the split line cutter portions remain secured to the vessel in an underwater environment.

12. The method of claim **8**, wherein the mounting bolt includes a pre-applied locking material retained in place by the bolt cavity, the method further comprising activating the locking material during step (e) to prevent the mounting bolt from loosening under vibration.

13. A kit for installing a split line cutter onto a marine propeller shaft, comprising:

- (a) a split line cutter including an upper portion, a lower portion, and at least one mounting bolt configured to secure the upper and lower portions around the propeller shaft;
- (b) an installation tool according to claim **1**, wherein the installation tool is sized to house and protect the split line cutter portions during storage and shipping; and
- (c) instructions describing a procedure for placing the split line cutter portions into the installation tool, aligning the tool on the propeller shaft, and tightening the mounting bolt using the installation tool, wherein the installation tool and the split line cutter are packaged and sold together for convenient, safe, and accurate installation.

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